Before and after: mathematical presentations

The same teaching content, revised for an audience

1. An overloaded page $\longrightarrow$ two pages with distinct jobs
2. An unexplained result figure $\longrightarrow$ an interpretable comparison
3. Repeated closing summaries $\longrightarrow$ one evidence-based conclusion

All examples use a standard quadratic problem. No private deck, author identity, or research measurement is reproduced.

Design note: Blue page headings identify the topic; BEFORE/AFTER labels identify the revision. Orange notes explain editing choices and are not part of a research talk.

1 / BEFORE: one page doing too many jobs

Original specimen: definition, algorithm, theorem, proof, interpretation

For $x \in \mathbb{R}^2$, $\kappa \geq 1$, $f_\kappa(x) = \frac{1}{2}(x_1^2 + \kappa x_2^2)$, $A = \operatorname{diag}(1, \kappa)$, $x_\star = 0$, $x_0 = (1,1)^T$, $\eta = 1/\kappa$:

$$\nabla f_\kappa = Ax, \quad x_{j+1} = (I - \eta A)x_j, \quad x_{1,j+1} = (1 - \kappa^{-1})x_{1,j}, \quad x_{2,j+1} = 0.$$

In general, for symmetric positive definite A with eigenvalues in $[m, L]$ and $0 < \eta < 2/L$, $\|x_j\|_2 \leq q^j \|x_0\|_2$ where $q = \max_i |1 - \eta\lambda_i| < 1$.

$$A = Q\Lambda Q^T, \quad Q^T Q = I, \quad Q^T x_{j+1} = (I - \eta\Lambda)Q^T x_j.$$

Orthogonal diagonalization gives independent scalar contractions; in our example $q = 1 - 1/\kappa$. Larger κ slows the first coordinate. For $j \geq 1$, $\|x_j\|_2 = (1 - 1/\kappa)^j$.

Design note: The mathematics is consistent, but the audience must learn the example and the general theorem at the same time. Smaller text is not the solution.

1 / AFTER A: define the example and expose the obstacle

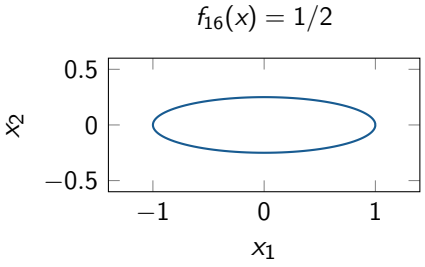
Job of this page: understand why one direction is slow

$$f_\kappa(x) = \tfrac{1}{2}(x_1^2 + \kappa x_2^2), \quad \kappa \geq 1.$$

Minimizer $x_\star = 0$; start $x_0 = (1, 1)^T$.

$$x_{j+1} = x_j - \kappa^{-1} \nabla f_\kappa(x_j).$$

$$x_{1,j+1} = (1 - \kappa^{-1}) x_{1,j}, \qquad x_{2,j+1} = 0.$$



Design note: Keep the problem and its mechanism together. Move the general theorem to the next page; explain the transition in the speaker notes.

1 / AFTER B: state the theorem and its decisive step

Job of this page: explain what makes the contraction valid

Let $A = A^T \succ 0$, with $0 < m \leq \lambda_i \leq L$. For $x_{j+1} = (I - \eta A)x_j$ and $0 < \eta < 2/L$,

$$\|x_j\|_2 \leq q^j \|x_0\|_2, \qquad q = \max_i |1 - \eta\lambda_i| < 1.$$

Key step: write $A = Q\Lambda Q^T$, with $Q^T Q = I$.

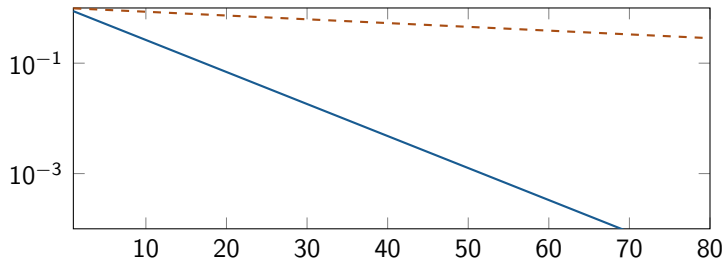
$$Q^T x_{j+1} = (I - \eta\Lambda) Q^T x_j.$$

Each component contracts by at most q; the norm is unchanged by Q. In the preceding example, $q = 1 - 1/\kappa$ and $\|x_j\|_2 = (1 - 1/\kappa)^j$ for $j \geq 1$.

Design note: Restore normal text size. Retain the hypotheses and the proof bridge; do not shorten the statement to an unconditional slogan.

2 / BEFORE: a figure without the experiment

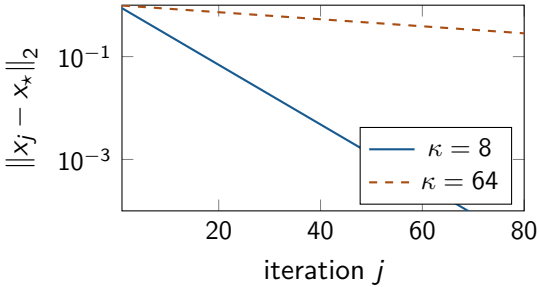
Original specimen: “Convergence results”



Design note: These curves are real formula evaluations, but their axes, parameters, and purpose are missing. A larger plot does not supply the missing meaning.

2 / AFTER: make the comparison interpretable

What changes: κ; what is measured: Euclidean error by iteration



Common problem

$$f_\kappa(x) = \tfrac{1}{2}(x_1^2 + \kappa x_2^2).$$

$x_0 = (1,1)^T$, $x_\star = 0$; gradient step $\eta = 1/\kappa$.

Exact reference

$$\|x_j\|_2 = (1 - 1/\kappa)^j, \quad j \geq 1.$$

Larger κ gives slower contraction under this step-size rule.

This is not a wall-clock comparison.

Design note: Keep the same curves. Add only the setup and labels needed to interpret them; use a line style as well as color.

3 / BEFORE: three closing pages repeat the same point

A storyboard of a repetitive ending; each box represents a page

Page A: Summary

Conditioning matters.

The spectrum controls the iteration.

Page B: Discussion

Large κ gives slow progress.

Directions behave differently.

Page C: Conclusion

The convergence rate depends on the eigenvalues.

Conditioning is important.

Design note: The statements are not false. The problem is repetition without another conclusion, useful limitation, or open question.

3 / AFTER: one closing page tied to the evidence

Replace three repetitive summaries with distinct supported statements

Conclusion	Support already shown
The step size must respect the spectrum.	$0 < \eta < 2/L$ ensures $\max_i \lvert 1 - \eta\lambda_i \rvert < 1$.
The selected step slows as κ grows.	In the example, $q = 1 - 1/\kappa$; the exact curves show this dependence.
No timing advantage has been established.	The comparison evaluates a recurrence, not solver wall-clock time.

Next question: what changes for a nonquadratic objective, where the curvature varies along the iteration?

Design note: Give the ending a new job: connect claims to evidence and state the next unresolved issue. Keep expanded discussion in the notes.

A reusable revision procedure

1. Identify the page's job and the audience's prerequisites.
2. Keep formulas and assumptions that carry that job.
3. Split, merge, or reorder to repair the dependency chain.
4. Add a figure only if it reveals a mechanism or comparison.
5. Check the revised pages, their transitions, and speaking time.

These are editing decisions, not mandatory page templates. A split is useful only if each resulting page has a distinct role; more slides alone do not make a better talk.

Design note: Use the companion guide for the edits, rationale, notes, and review criteria. This final page belongs to the demonstration, not the toy research talk.